

Graph the functions, draw a sample of a cross section of an arbitrary shell, and then compute. (Hint: You will sometimes need to use multiple integrals.)

1. Use the shell method to find the volume of the solid obtained by revolving the region enclosed by $y = \sqrt{x+1}$, $y = 1$, $x = 0$, and $x = 3$ about the y -axis.
2. Use the shell method to find the volume of the solid obtained by revolving the regions enclosed by $y = x^2$, $y = 2 - x$, $x = 0$, and $x = 2$ about the y -axis.

3. Use the shell method to find the volume of the solid obtained by revolving the region enclosed by $y = 1 + \sqrt{x}$, $y = 1 - \sqrt{x}$, $x = 0$, and $x = 1$ about the x -axis.

4. Use the shell method to find the volume of the solid obtained by revolving the regions enclosed by $y = x + 1$, $y = 2 - x$, and $x = 0$ and $x = 2$ about the line $y = -1$.

5. Now use a combination of disks/washers and shells. Find the volume of the solid obtained by revolving the regions enclosed by $y = x + 1$, $y = 2 - x$, and $x = 0$ and $x = 2$ about the y -axis. A cross-section of this solid should be a wonky bow tie looking thing.

Use disks for the 'left' piece (inner solid) and shells for the 'right' piece (outer solid) or shells for the 'left', and washers for the 'right'.